Graphcore Intelligence Processing Unit (IPU) v1/v2

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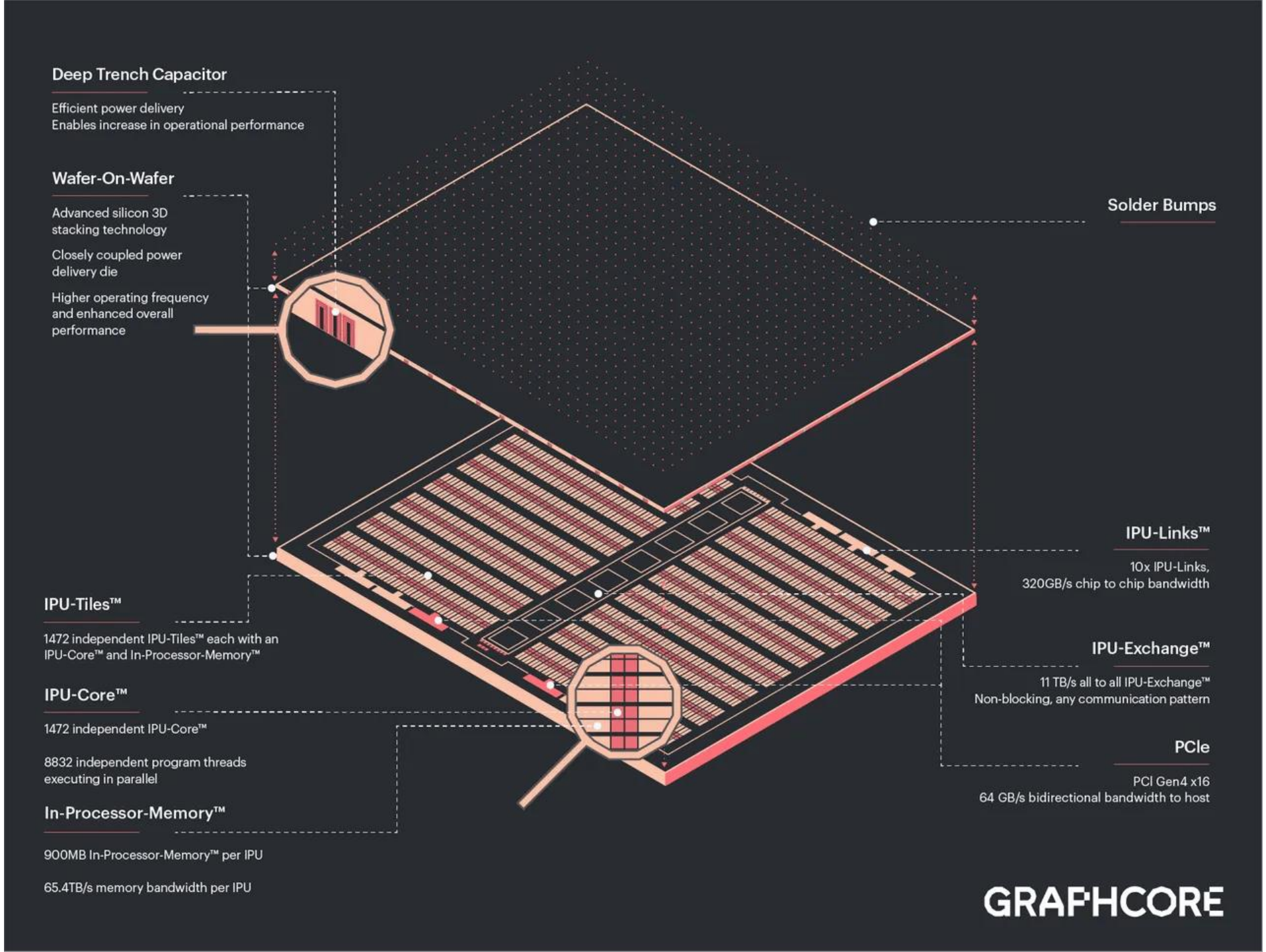
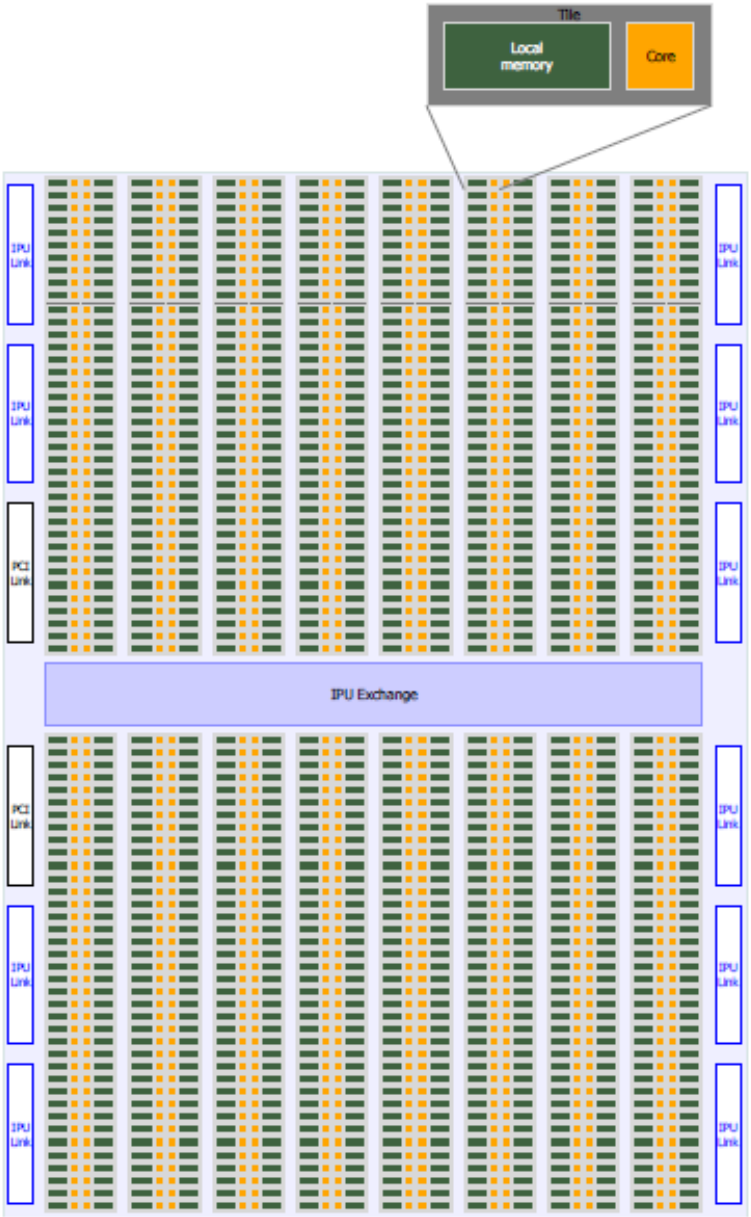
Courtesy: Graphcore website and papers


Deep Trench Capacitor
Efficient power delivery
Enables increase in operational performance
Wafer-On-Wafer
Advanced silicon 3D stacking technology
Closely coupled power delivery die
Higher operating frequency and enhanced overall performance
Solder Bumps
IPU-Links™
10x IPU-Links, 320GB/s chip to chip bandwidth
IPU-Tiles™
1472 independent IPU-Tiles™ each with an IPU-Core™ and In-Processor-Memory™
IPU-Exchange™
11 TB/s all to all IPU-Exchange™
Non-blocking, any communication pattern
IPU-Core™
1472 independent IPU-Core™
8832 independent program threads executing in parallel
PCIe
PCI Gen4 x16
64 GB/s bidirectional bandwidth to host
In-Processor-Memory™
900MB In-Processor-Memory™ per IPU
65.4TB/s memory bandwidth per IPU
GRAPHCORE

Architecture



1,216 tiles, each with one core and 256KB local memory

Distributed, local memories

- IPU memories are scratchpads, not caches. No shared memory
- Memory is SRAM with 45 TB/s aggregate bandwidth and 6 cycle latency
- IPU uses MIMD
- If working set fits in local memory, there is no cost of control-flow divergence.
- This is because each core accesses local memory at a fixed cost, independent of access patterns.
- Thus, IPU is more efficient than GPUs at applications with irregular memory accesses or control-flow dominated.
- Graphcore’s design uses local memory instead of memory like HBM2(e) as NVIDIA/Intel server GPUs and other accelerators use. This trades capacity for bandwidth and latency due to the proximity.

Multithreading

- Similar to CPU and GPU, IPU achieves higher efficiency by oversubscribing threads to cores.
- Each IPU tile maintains 6 threads (Similar to Intel's hyper-threading).
- This hides instruction execution latencies.
- Each tile rotates among threads according to a static, round-robin schedule.

Core architecture

- Each IPU tile has Accumulating Matrix Product (AMP) units.
- AMP units accelerate matrix multiplication and convolution.
- An AMP unit can finish 64 mixed-precision or 16 single-precision FLOPs per clock cycle.
- At a 1.6-GHz clock rate, the 1,216 tiles on one IPU deliver 31.1 and 124.5 TFlops/s in single and mixed precision, respectively

- AMP units are similar in purpose to the GPU's TensorCore units
- IPU does NOT support double-precision.

GPU vs IPU

Arithmetic		Graphcore C2 IPU		NVidia Volta V100 (SXM2)	
Precision		Units	1 IPU	Units	1 GPU
Single (FP32)	Theoretical	AMP	31.1	TensorCores	N/A
	Theoretical	Vector	7.8	FP cores	15.7
	Actual GEMM	AMP	18.9	FP cores	15.5
	% Theor.		60.7%		98.7%
Mixed (FP16.32)	Theoretical	AMP	124.5	TensorCores	125.0
	Theoretical	Vector	15.6	FP cores	31.4
	Actual GEMM	AMP	58.9	TensorCores	90.0
	% Theor.		47.3%		72.0%

GPU vs IPU

- **Single precision results**:
- one IPU processor offers ~2X throughput as one V100 GPU: 31.1 vs. 15.7 TFlops/s.
- Reason: On GPU, TensorCores do not support single precision and hence, regular FP cores are used.
- **Mixed precision results**:
- one IPU processor matches one V100 GPU (124.5 vs. 125.0 TFlops/s).

GPU vs IPU

- IPU delivers higher throughput,
- GPU supports larger operands thanks to its higher device memory capacity.
- The largest square matrix operands fitting one IPU are 2,944*2,944, while on a 32-GB GPU they are roughly 50,000 *50,000

- Each IPU uses ~150W of power
- NVIDIA V100 has 250W TDP

Memory hierarchy comparison

- The IPU's local memory has lower latency than the fastest memories on the Turing GPUs and is on par with the L2 cache in modern Intel CPUs.
- However, IPU's local memory is vastly larger than those memories in a per-chip comparison.

Architecture	Memory	Per-chip Capacity (MiB)	Latency (ns)	Latency (cycles)	Clock Frequency (GHz)
Graphcore IPU	Tile-local	304	3.75	6	1.60
NVidia T4 GPU	Shared L1	1.25 ... 2.5 1.25 ... 2.5	11.94 20.13	19 32	1.59
Intel *-Lake CPU	L1D L2	0.25 ... 0.875 4 ... 28	0.93 ... 1.92 2.79 ... 4.62	4 ... 5 12	2.60 ... 4.30

Bandwidth comparison

TABLE II: Comparison of IPU, GPU and CPU memory hierarchies (running STREAM as per [18])

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Platform	Memory	Size	Bandwidth
			GB/s
IPU	SRAM	304 MiB	7,261 .. 12,420
GPU: NVIDIA V100	L1/shared	10 MiB	11,577
	L2	6 MiB	3,521
	HBM-2	16 GiB	805
CPU: Intel Skylake (2x 24 cores)	L1	1.5MiB	7,048
	L2	48 MiB	5,424
	L3	66 MiB	1,927
	DRAM	768 GiB	225

Scalability

- Software designers must partition their working set across the tiles' memories appropriately and make tiles exchange data with each other when they need non-local operands.
- Tiles can exchange data using a very fast all-to-all, on-chip interconnect (theoretically 8 TB/s)
- Total memory = 304 MB per IPU (608 MB per board)
- A model requiring more than this can be divided across IPU processors and IPU boards

GC2 “Colossus Mk1” IPU [2018 power-on]

23,647,173,309 active transistors in TSMC N16

1216 processor tiles @ 256KiB

Total 125Tflop/s + 304MiB SRAM

62TB/s memory, 7.8TB/s inter-tile, 320GB/s inter-chip

GC200 “Colossus Mk2” IPU [2020 power-on]

59,334,610,787 active transistors in TSMC N7

1472 processor tiles @ 624KiB

Total 250Tflop/s + 896MiB SRAM

62TB/s memory, 7.8TB/s inter-tile, 320GB/s inter-chip

https://www.servethehome.com/graphcore-bow-ipu-launched/

M2000

IPU-MACHINE

1 PetaFlop IPU compute
2.8Tbps IPU-Fabric™
$32,450 *MSRP*

M2000 PRODUCT CONFIGURATIONS

1x M2000

4x M2000

8x M2000

DIRECT
ATTACHED

IPU-POD$_6$

COLOSSUS MK2

the worlds most complex processor

59.4Bn transistors, TSMC 7nm @ 823mm^2

250TFlops AI-Float | 900MB In-Processor-Memory

1472 independent processor cores

8832 separate parallel threads

>8x step-up in system performance vs Mk1

GC200 - IPU

LATEST GPU vs. COLOSSUS Mk2 IPU

	NVIDIA DGX-A100 (8x A100)	GRAPHCORE 8x M2000	
FP32 compute	156TFLOP	2PFLOP	**>12x**
AI compute	2.5PFLOP[1]	8PFLOP[2]	**>3x**
AI Memory	320GB[3]	3.6TB[4]	**>10x**
System Price	$199,000$_{MSRP}$	$259,600$_{MSRP}$	

NOTES:
[1] Actual figure for TF32/FP16. NVIDIA 8xA100 5PFlop reference is for 50% sparsity which includes Pflops for operations that aren't run
[2] Graphcore AI Float with IEEE FP16.16 multiply.accumulate and IEEE FP16.SR 16bit float with stochastic rounding, with equivalent accuracy performance as FP32
[3] 40GB HBM memory on A100 modules *8 modules per DGX-A100 system
[4] IPU-Exchange Memory which includes attached DRAM and IPU In-Processor-Memory with 100x bandwidth vs. HBM memory sub-system

INTRODUCING THE BOW IPU
WORLD'S FIRST 3D WAFER-ON-WAFER PROCESSOR

3D silicon wafer stacked processor

350 TeraFLOPS AI compute

Optimized silicon power delivery

0.9 GigaByte In-Processor-Memory @ 65TB/s

1,472 independent processor cores

8,832 independent parallel programs

10x IPU-Links™ delivering 320GB/s

GRAPHCORE

References

- Dissecting the Graphcore IPU Architecture via Microbenchmarking
- Using the Graphcore IPU for traditional HPC applications
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- https://www.graphcore.ai/posts/introducing-second-generation-ipu-systems-for-ai-at-scale
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